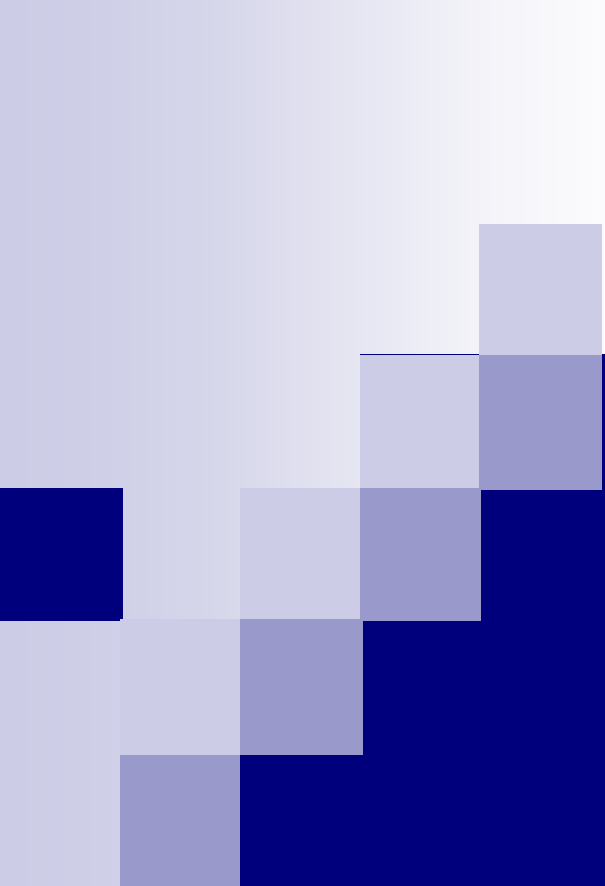




BIOAVAILABILITY AND BIOEQUIVALENCE

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OUTLINE

- **INTRODUCTION**
- **RELATIVE AND ABSOLUTE BIOAVAILABILITY**
- **OBJECTIVES OF BIOAVAILABILITY STUDIES**
- **METHODS OF ASSESSING BIOAVAILABILITY**
- **BIOEQUIVALENCE STUDIES**

INTRODUCTION

- With increasing availability and use of generic drug products, health care professionals are confronted with ever-larger array of multisource products from which they must select those that are *therapeutically equivalent*.
 - ✓ Makes these issues very relevant to pharmacists in all settings
- After patent for brand-name drug expires, pharmaceutical firm may manufacture generic drug product that can be substituted for branded drug product
 - ✓ Generic drug manufacturer must demonstrate that generic drug product is *bioequivalent* and *therapeutically equivalent* to brand name product

INTRODUCTION...

Pharmaceutical equivalents

- Contain same amount of active substance(s), i.e., same salt or ester form, in same DF that meet same or comparable standards (i.e., strength, quality, purity, and identity)
- May differ in excipients (including colors, flavors, preservatives), characteristics such as shape, scoring configuration, release mechanisms, packaging, expiration time, and, within certain limits, labeling

INTRODUCTION...

Pharmaceutical alternatives

- Contain same therapeutic moiety but as different salts, esters, or complexes

E.g., TTC PO₄ and TTC HCl equivalent to 250 mg base

- Different DFs and strengths within product line by single manufacturer are pharmaceutical alternatives

E.g., Extended-release DF and std Immediate-release DF of same active ingredient

Tablet and Capsule DFs containing same active ingredient

Bioavailability

- Rate and extent to which active ingredient is absorbed from pharmaceutical product and becomes available at site of action

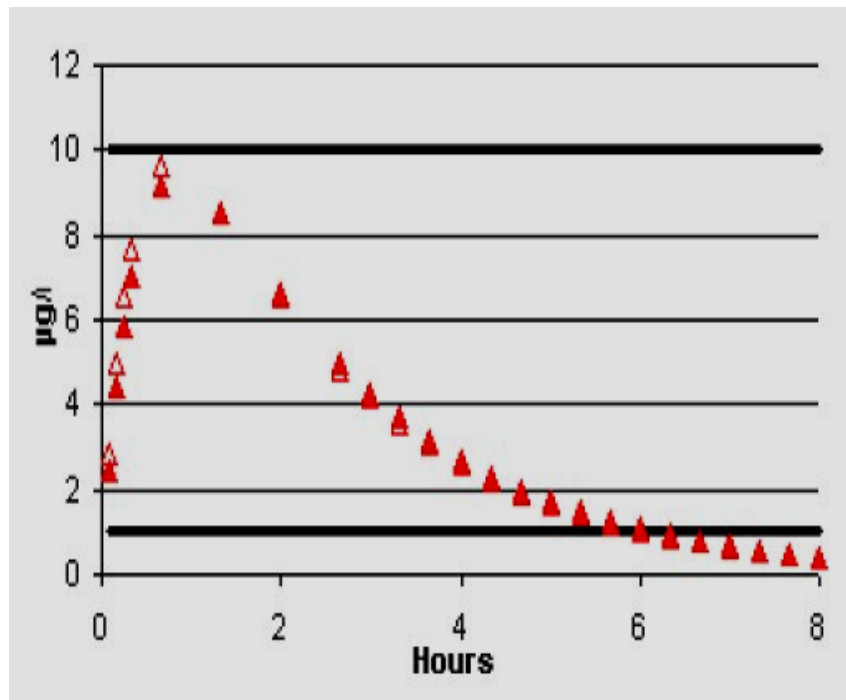
INTRODUCTION...

Bioequivalents

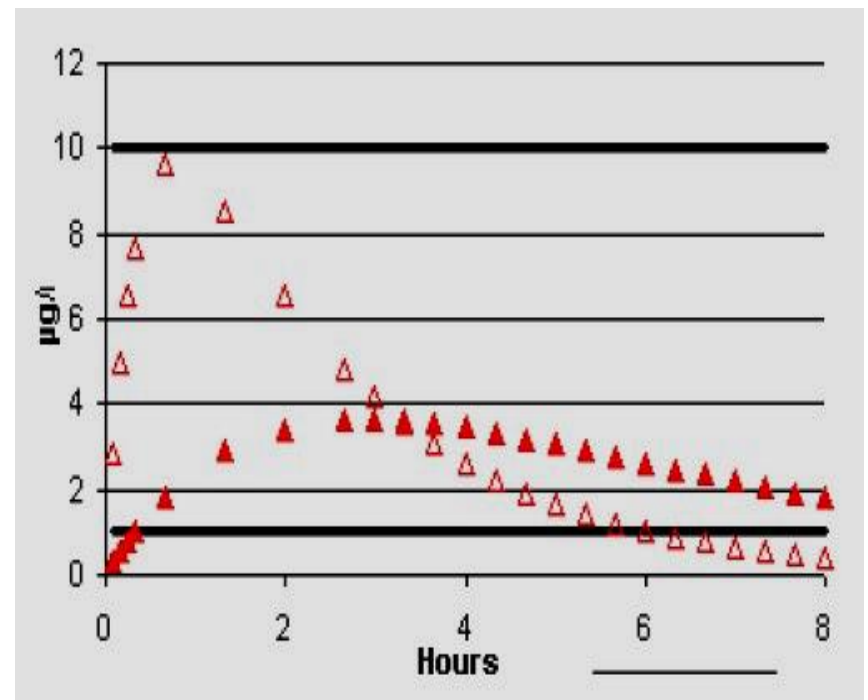
- Describes pharmaceutical equivalent or pharmaceutical alternative products that display comparable bioavailability when administered in same molar dose to such degree that their effects w.r.t. both efficacy and safety is essentially same
- Bioequivalence may sometimes be demonstrated using *in vitro* bioequivalence standard, especially when such *in vitro* test has been correlated with human *in-vivo* bioavailability data
 - *in vitro/in vivo* correlation

INTRODUCTION...

Two DFs are
bioequivalent



Two DFs are not
bioequivalent



INTRODUCTION...

Therapeutic equivalents

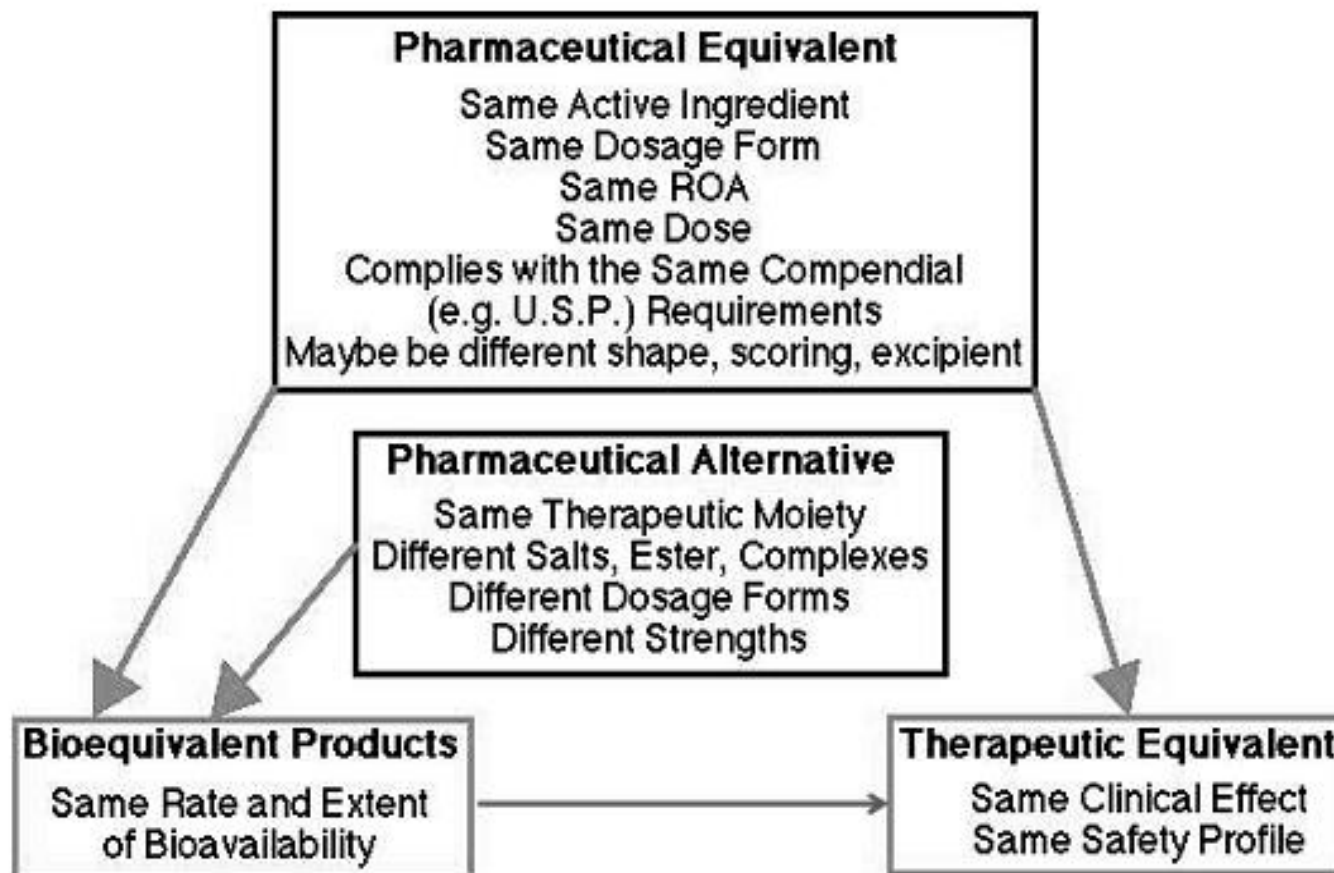
- Contain same therapeutically active drug that should give same therapeutic effect and have equal potential for adverse effects under conditions set forth in the labels of these products
- Therapeutic equivalent drug products must be (FDA)
 - (1) Approved as safe and effective
 - (2) Pharmaceutical equivalents
 - (3) Bioequivalent
 - (4) Adequately labeled; and
 - (5) Manufactured in compliance with cGMP regulations

INTRODUCTION...

Therapeutic alternatives

- Contain different active ingredients that are indicated for same therapeutic or clinical objectives
- Active ingredients are from same pharmacologic class and are expected to have same therapeutic effect when administered to patients
 - Ibuprofen is given instead of Aspirin
 - Cimetidine may be given instead of Ranitidine

INTRODUCTION...



Relative and Absolute Bioavailability

- Bioavailability of drug from DF is critical element of drug product's clinical efficacy
 - pharmacologic response is generally related to conc. of drug at receptor site
- Most bioavailability studies involve determination of drug conc. in **blood** or **urine**
 - ✓ Indirect measure of drug response by monitoring level in blood or urine

Relative and Absolute...

- Bioavailability concerns with *fraction of dose* that actually reaches bloodstream after specific dose is administered
- ***"Absolute" bioavailability***, F , is fraction of administered dose which actually reaches systemic circulation
 - ***Generally less than amount of drug actually administered***
 - ***$F = 0$ (no drug absorption) to $F = 1$ (complete absorption)***
- Absolute bioavailability of drug is systemic availability of drug after extravascular administration (e.g., oral, rectal, subcutaneous) compared to *IV route*

Relative and Absolute...

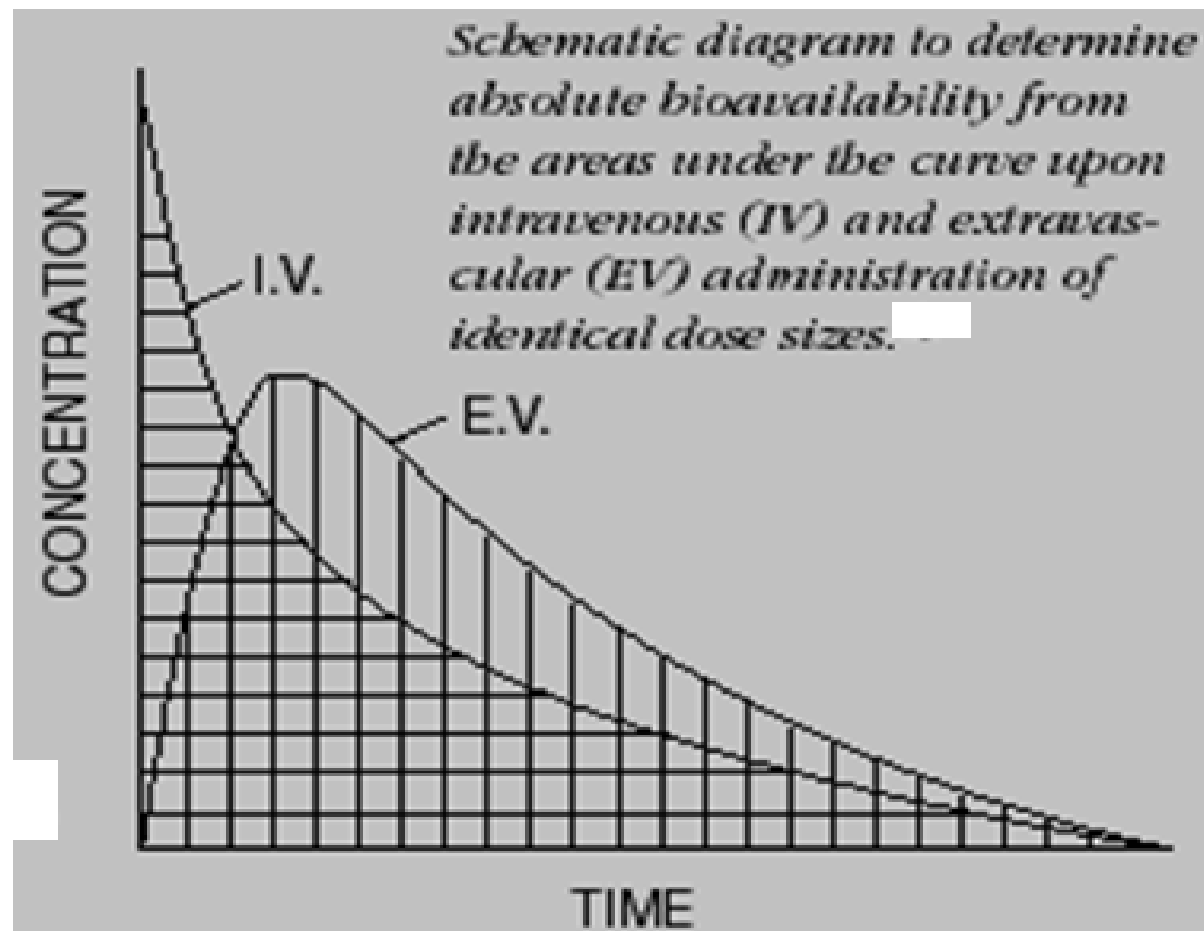
- Absolute bioavailability after oral drug administration using plasma data can be determined as follows:

$$\text{Absolute Bioavailability} = F = \frac{AUC_{PO}}{AUC_{IV}}$$

- Considering possibility of different doses:

$$\text{Absolute Bioavailability} = F = \frac{AUC_{PO} / \text{Dose}_{PO}}{AUC_{IV} / \text{Dose}_{IV}}$$

Relative and Absolute...



Relative and Absolute...

- Urinary drug excretion data may also be used to measure bioavailability
 - ✓ total amount of intact drug excreted in urine is collected

$$\text{Absolute Bioavailability} = F = \frac{\frac{[D_u]_{PO}^{\infty}}{\text{Dose}_{PO}}}{\frac{[D_u]_{IV}^{\infty}}{\text{Dose}_{IV}}}$$

Where

- ✓ $[D_u]^{\infty}$ is total amount of drug excreted in urine
- ✓ PO is oral or any other extravascular route

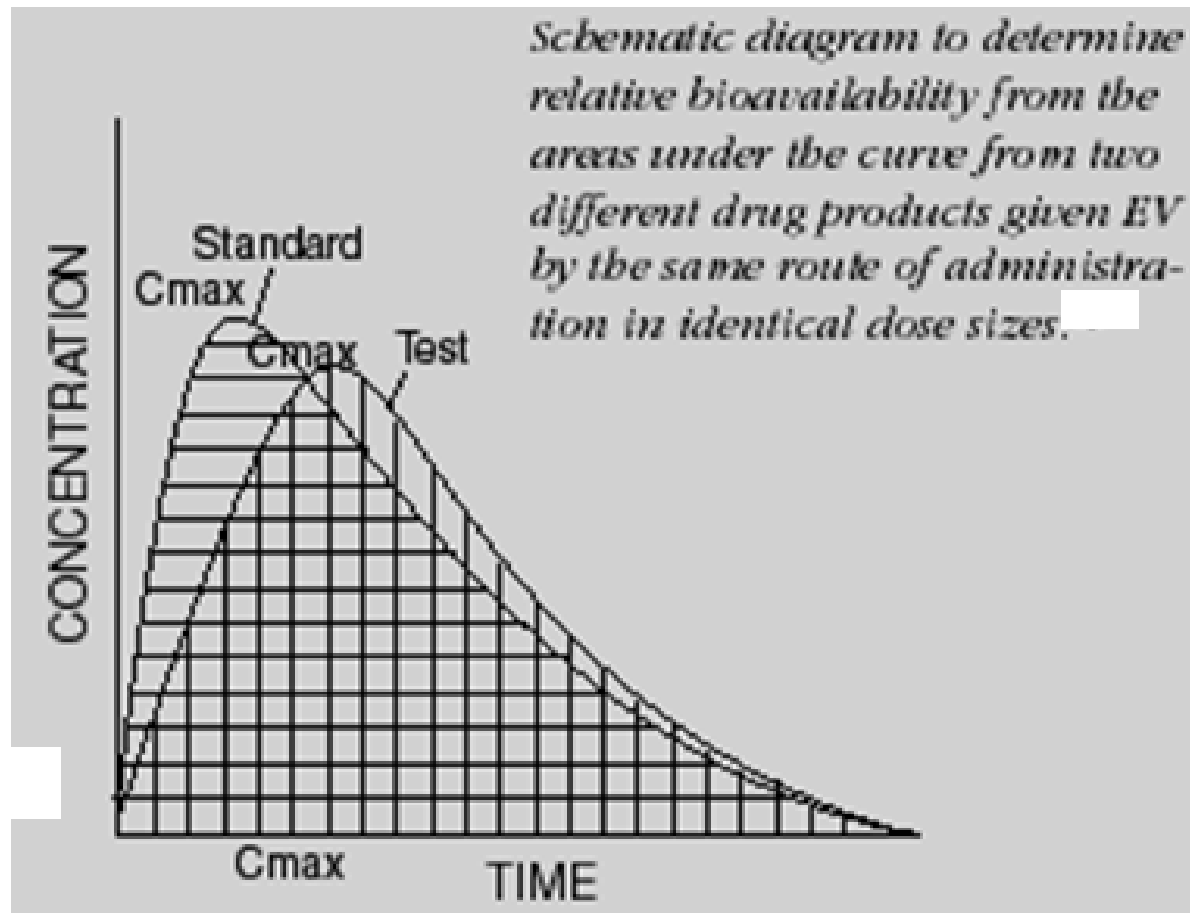
Relative and Absolute...

- **"Relative" bioavailability** refers to availability of drug product as compared to a recognized standard
- Relative bioavailability of **product A** compared to **product B**, is given by:

$$\text{Relative Bioavailability} = \frac{\text{AUC}_A / \text{Dose}_A}{\text{AUC}_B / \text{Dose}_B}$$

where drug **product B** is reference/recognized standard

Relative and Absolute...



Relative and Absolute...

- The percent relative bioavailability using urinary excretion data can be determined as follows:

$$\text{Percent Relative Bioavailability} = \frac{[D_u]_A^{\infty}}{[D_u]_B^{\infty}} \times 100$$

where

- ✓ $[D_u]^{\infty}$ is total amount of drug excreted in urine
- ✓ $F_{rel} = 0$ (no drug absorption) to $F_{rel} = 1$ (equal absorption) to $F_{rel} > 1$ (more drug absorption)

Relative and Absolute...

Example:

- ✓ The BA of a new investigational drug was studied in 12 volunteers. Each volunteer received either a single oral tablet containing 200 mg of the drug, 5 mL of a pure aqueous solution containing 200 mg of the drug, or a single IV bolus injection containing 50 mg of the drug. Plasma samples were obtained periodically up to 48 hours after the dose and assayed for drug concentration. The average AUC values (0–48 hours) are given in the table below.

Relative and Absolute...

From these data, calculate

1. The **relative BA** of the drug from the oral solution compared to the tablet and
2. The **relative BA** of the drug from the tablet compared to oral solution
3. The **absolute BA** of the drug from the tablet.

Drug product	Dose (mg)	AUC ($\mu\text{g hr/ml}$)
Oral tablet	200	86.1
Oral solution	200	89.5
IV bolus injection	50	37.8

Objectives of Bioavailability studies

- BA studies are important in the:-
 1. Primary stages of development of a suitable DF for a new drug entity
 2. Determination of influence of excipients, patient related factors and possible interaction with other drugs on the efficiency of absorption
 3. Development of new formulations of the existing drugs
 4. Control of quality of a drug product during the early stages of marketing
- Determine the influence of processing factors, storage and stability on drug absorption.

Methods of Assessing Bioavailability

- Different methods are used to assess drug bioavailability

** Plasma drug concentration data*

- Time for peak plasma concentration ($t_{\max}$)
- Peak plasma drug concentration ($C_{\max}$)
- Area under plasma drug concentration–time curve (AUC)

** Urinary drug excretion data*

- Cumulative amount of drug excreted in urine (D_u^{∞})
- Rate of drug excretion in urine (dD_u/dt)
- Time for maximum urinary excretion (t^{∞})

Methods of Assessing...

* *Acute pharmacodynamic effect*

- Maximum pharmacodynamic effect
- Time for maximum pharmacodynamic effect
- Area under pharmacodynamic effect-time curve
- Onset time for pharmacodynamic effect

* *Clinical observations*

- Well-controlled clinical trials

* *In vitro studies*

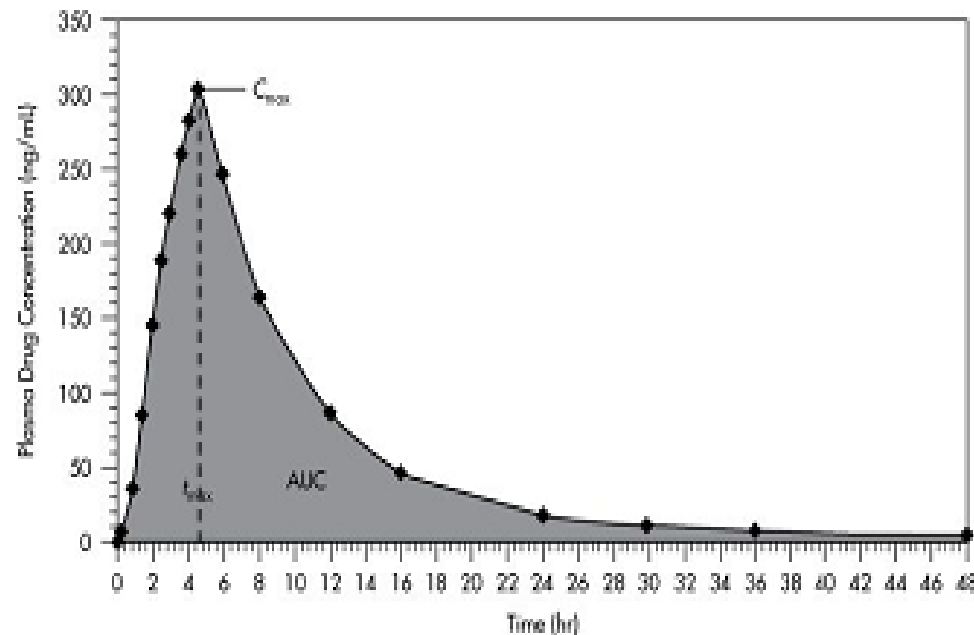
Methods of Assessing...

Plasma drug concentration data

- Most widely used and accepted method for assessing bioavailability
- By definition, bioavailability is measure of **rate** and **extent** of absorption of drug from drug product

- $t_{\max}$ and $C_{\max}$ are measures of rate of absorption

-**AUC** is measure of extent of absorption



Methods of Assessing...

t_{max} -time required to reach maximum drug conc.

- **At t_{max} ,**
 - Peak drug absorption occurs
 - Rate of absorption exactly equals rate of drug elimination
 - Drug absorption still continues after t_{max} is reached, but at slower rate
- **t_{max} can be used as approximate indication of absorption rate**
 - t_{max} become smaller as absorption rate becomes more rapid

Methods of Assessing...

C_{max} - *maximum plasma drug conc.*

- For many drugs, relationship is found between plasma drug conc. and pharmacodynamic effect
 - C_{max} provides indications that drug is sufficiently systemically absorbed to provide therapeutic response
 - C_{max} provides warning of possibly toxic levels of drug
- C_{max} is often used in bioequivalence studies as surrogate measure for rate of drug bioavailability
 - **$Rate \uparrow \rightarrow C_{max} \uparrow$**

Methods of Assessing...

AUC-measurement of extent of drug bioavailability

- Reflects total amount of active drug that reaches systemic circulation
- AUC is area under plasma level-time curve from $t = 0$ to $t = \infty$,

$$AUC = [AUC]_0^t + [AUC]_t^{\infty}$$

where

- t is last time point of plasma sample collection

- $[AUC]_0^t$ is calculated by trapezoidal rule;

$$\text{Area} = (1/2) (C_1 + C_2) (t_2 - t_1) + (1/2) (C_2 + C_3) (t_3 - t_2) + \dots + (1/2) (C_{n-1} + C_n) (t_n - t_{n-1})$$

Bioequivalence Studies

- Substitution of one product for another can be made provided they are equally effective therapeutically as the standard accepted.
- In order to ensure clinical performance of such drug products, bioequivalence studies should be performed.
- As the BA studies, either plasma level or urinary excretion studies may be performed to assess bioequivalence between drug products.

Bioequivalence Studies . . .

Study design

A. Parallel design

- Two formulations are administered to two groups of volunteers
- To avoid bias, formulations may be administered randomly to volunteers
- **Disadvantage:** inter-subject variation isn't being corrected
 - Most of the time, inter-subject variation is greater than variation between formulations
- Used if half-life of drug is very long
- Substantially greater number of subjects are needed

Bioequivalence Studies . . .

B. Cross-over design

- Minimizes substantial inter-subject variability in study by using each subject as his own control
- Types of cross-over designs:
 1. **Complete / Latin square cross-over design**
 2. **Balanced incomplete block design (BIBD)**

Bioequivalence Studies . . .

1. Latin square cross-over design

- Each subject receives each formulation only once with adequate time between medications for elimination of drug
- Complete cross-over study of two products A and B in 12 subjects

Group	Subjects	Period 1	Period 2
1	1 - 6	A	B
2	7 - 12	B	A

- *Period*: time period in which a study is performed
- *Sequence*: number of different orders of treatment groups
 - two-sequence, two-period study

Bioequivalence Studies...

Advantages

1. Minimizes inter-subject variability in study

→ Each subject serves as his own control

2. Minimizes carryover effect

→ All patients don't receive same drug product on same day and in same order

→ Variations due to sequence, period or treatment are reduced

→ Each formulation is preceded and succeeded by the other formulations

3. Requires less number of subjects to get meaningful results

Bioequivalence Studies . . .

Disadvantages

1. Requires longer time to complete since wash out period exists between study periods
→ Longer biological half-life
 2. Time to complete trial depends on number of formulations evaluated in study
 3. Increased number of study periods leads to high subject dropouts
 4. Medical ethics doesn't allow too many trials on a subject continuously for longer time
- * Disadvantages are overcome by using BIBD

Bioequivalence Studies . . .

2. Balanced incomplete block design (BIBD)

1. Each subject receives not more than two formulations
2. Each formulation is administered same number of times
3. Each pair of formulation occurs together in same number of subjects

* BIBD for four formulations A, B, C, D in 12 subjects

Bioequivalence Studies...

- Each subject receives two formulations
- Each formulation is administered six times
- Each pair of formulations occurs together in two subjects (AB, AC, AD, BC, BD, CD)

Subject	Treatment period	
	I	II
1	A	B
2	B	A
3	A	C
4	C	A
5	A	D
6	D	A
7	B	C
8	C	B
9	B	D
10	D	B
11	C	D
12	D	C

Bioequivalence Studies . . .

- The data in represent the average findings in antibiotic plasma samples taken from 10 humans (average weight 70 kg), tabulated in a four-way crossover design.
- A. Which of the four drug products in would be preferred as a reference standard for the determination of relative bioavailability? Why?
- B. From which oral drug product is the drug absorbed more rapidly?
- C. What is the absolute BA of the drug from the oral solution?
- D. What is the relative BA of the drug from the oral tablet compared to the reference standard?

	Plasma Concentration ($\mu\text{g/ml}$)			
Time after Dose (hr)	IV Solution (2 mg/kg)	Oral Solution (10 mg/kg)	Oral Tablet (10 mg/kg)	Oral Capsule (10 mg/kg)
0.5	5.94	23.4	13.2	18.7
1.0	5.30	26.6	18.0	21.3
1.5	4.72	25.2	19.0	20.1
2.0	4.21	22.8	18.3	18.2
3.0	3.34	18.2	15.4	14.6
4.0	2.66	14.5	12.5	11.6
6.0	1.68	9.14	7.92	7.31
8.0	1.06	5.77	5.00	4.61
10.0	0.67	3.64	3.16	2.91
12.0	0.42	2.30	1.99	1.83
$\text{AUC} \left(\frac{\mu\text{g}}{\text{mL}} \times \text{hr} \right)$	29.0	145.0	116.0	116.0